

AURIX

Microcontroller Infineon Software Architecture

User's Manual

Multi-core Support

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Document Change History

Date	Version	Changed By	Change Description
2013-04-29	1.0	Angeswaran Thangaswamy	Initial version for Aurix

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Table of Contents		Page
1	MCAL and multi-core	6
1.1	MCU, PORT	6
1.2	DIO	6
1.3	WDG.....	6
2	Allocating a driver to core	6

List of Tables

Page

No table of figures entries found.

1 MCAL and multi-core

This document explains on multi-core support of AURIX MC-ISAR project. All MCAL drivers can be executed from any core. The given demo application is associated with core 0 for all drivers. A start up file (Mcal.c) is given as evaluation to start and run other cores.

AUTOSAR EcuM recommends executing all the MCAL drivers from one core. However by design of the driver it may be possible to run some drivers from multiple cores. The following points shall be considered before designing the application

- The initialization of the driver shall be done only once and from one core only.
- The allocation of global variables is done in only one section and it is continuous. Assume ADC variables are placed in DSPR (Data Scratch Pad RAM) of core 0 and Adc_StartGroupConversion() of a particular group is called from core 1 then the RAM associated of that group is still placed in DSPR of core 0 and some more cycles are required to access the same.

It is recommended to execute MCAL from one core only. The following drivers can undergo parallel execution.

1.1 MCU, PORT

The driver's main work is only with initialization and parallel execution doesn't apply here.

1.2 DIO

It is possible to do parallel execution provided the ports accessed by both cores are independent.

1.3 WDG

Each CPU has its own watchdog timer and the design is done to suit the same. Parallel execution is possible.

2 Allocating a driver to core

The following section explains how to allocate ADC to core 1.

Step 1: Allocation of variables of ADC

ADC variables are preferred to be placed in DSPR memory of CORE 1 (Private variables). If required, ADC variables can be placed in LMU Ram memory as well (Shared variables).

To place the variable in DSPR of CORE 1, the following has to be done

Eg:

```
#define ADC_START_SEC_VAR_16BIT
```

```
#include "MemMap.h"
```

```
static uint16 Adc_AllRunningCh[ADC_KERNEL_USED_COUNT];
```

```
#define ADC_STOP_SEC_VAR_16BIT
```

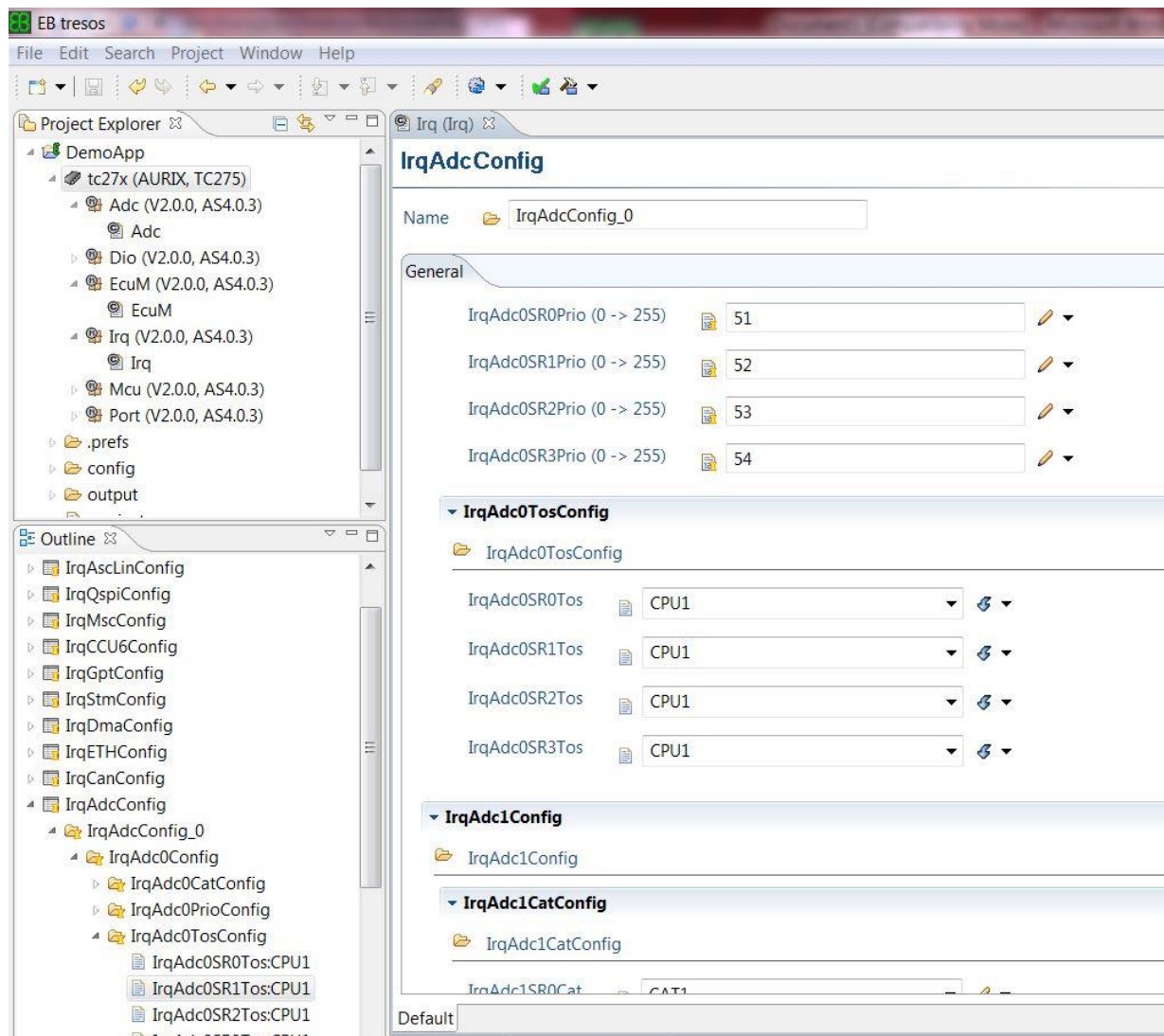
```
#include "MemMap.h"
```

- a) The section ADC_START_SEC_VAR_16BIT is redefined as DEFAULT_START_SEC_VAR_16BIT_SHARED in MemMap.h. Change the DEFAULT_START_SEC_VAR_16BIT_SHARED to DEFAULT_START_SEC_VAR_16BIT_CORE1_PRIVATE.
- b) The section ADC_STOP_SEC_VAR_16BIT is redefined as DEFAULT_STOP_SEC_VAR_16BIT_SHARED in MemMap.h. Change the

DEFAULT_STOP_SEC_VAR_16BIT_SHARED to
DEFAULT_STOP_SEC_VAR_16BIT_CORE1_PRIVATE.

Step 2: Mapping of ADC interrupt service routines to core 1

In Irq module configuration, select CPU1 as the Type of Service of all the required ADC interrupts.



Step 3: Start and initialize core 1 from start-up code.

In our sample code (Mcal.c) the same is done by function `Mcal_StartCore()`. At the end of initialization (Stack Pointer, CSA etc) branch it to a particular function where core 1 API calls are made. The same is given in function `Mcal_Core1Container()`.

Step 4: Initialize ADC SW driver.

Initializing through EcuM for all drivers in core 0 and only ADC in core 1 is not possible. As per AUTOSAR EcuM all SW modules has to be initialized from one core. However to support `Adc_Init()` from core 1, the following manual modifications are required

- Remove ADC initialization from EcuM (i.e. in tresos configuration)
- Include `Adc.h` in `Mcal.c`

- c) ECUM_USES_ADC has to be defined in either EcuM_Cfg.h / can be defined via compiler option.
- d) Call Adc_Init(&Adc_ConfigRoot[0]) from Mcal_Core1Container() function. (Before calling ensure that modules like MCU is initialized and verified through some semaphore)

Step 5: Executing the functionality

Run required function from the base task of Mcal_Core1Container().

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